

Emergence Of SUPERCONDUCTING ELECTRONICS



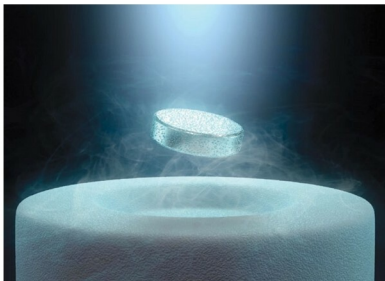
Dr S.S. Verma
is a professor
at Department
of Physics, Sant
Longowal Institute
of Engineering and
Technology, Sangrur,
Punjab

The electrical resistivity of a metallic conductor decreases gradually as its temperature is lowered. In ordinary conductors, this decrease is limited by impurities and other defects. So even near absolute zero, normal conductor shows some resistivity. On the other hand, in a superconductor, the resistance drops abruptly to zero when the material is cooled below its critical temperature. Thus, an electric current flowing through a loop of superconductor can persist indefinitely with no power source.

Superconductivity has been instrumental in bagging *The Noble Prize in Physics* five times for findings as listed in the table on next page.

Superconductors are touted as one of the greatest discoveries of quantum physics. These are already being used for magnets in MRI machines, levitation transport and many other applications. Superconductors may one day fuel superfast computers, which will, of course, benefit intelligence agencies that need to process and analyse vast quantities of data. Superconducting technologies may also pave the way for superior electrical grids and even magnetic levitation trains.

Magnetic levitation



Superconducting electronics

Though low- to high-temperature superconducting materials have their own applications on the basis of superconductivity alone, semiconductors have a decade-long headstart and dominate the electronics industry as well. For a long time, scientists have talked about whether superconductors would be a viable alternative to semiconductors for advanced computing as superconductor enable lower-power, faster, radiation-hardened electronics.

Further, with a growing demand for electronics having miraculous characteristics (low noise, low loss, low dissipation, less weight, high resolution, high speed, high frequency, etc), use of electronic and superconductivity properties of materials—together known as ‘superconducting electronics’ or ‘electronics with superconductors’—is turning to be a field with varied applications, including superconducting circuits, superconducting quantum interference devices (SQUIDS) magnetometers, current limiters and electronic filters. Superconducting electronics is of great interest for several niche applications, like ultrafast routers for communication networks, analogue-to-digital converters working in the microwave field and ultra-sensitive digital receivers, which have no counterpart in the semiconductor world.

Scientists across the world are investigating superconducting materials that could transform electronics in much the same way as silicon-based semiconductors did. Superconducting materials offer less resistance than semiconductors, so currents flow much faster. Researchers intend to build software applications that will make it easier to design and develop superconducting networks to power future supercomputers capable of much faster processing with lower energy requirements.